Prayush Desai

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EDUCATION

University of Wisconsin–Madison Madison, WI
Bachelor of Science in Computer Science | GPA: 3.58/4.00 *Expected May 2029*
Relevant Coursework: Data Structures & Object-Oriented Programming in Java, Calculus I & II | **In progress:** Programming III, Discrete Mathematics, Intro to Computer Engineering, Data Science Modeling I

PROJECTS

Smart Scheduler | Python, Flask, scikit-learn, NumPy, Jinja2, HTML May 2026 – Present
- Built a Flask web app where students enter courses with multiple lecture and discussion sections, then receive every conflict-free weekly schedule, ranked
- Generated schedules by pairing each lecture with its linked discussions and taking the Cartesian product across courses, filtering with pairwise day-and-time overlap checks
- Designed a 6-factor scoring function that ranks schedules on credit load, day spread, early and evening classes, and gaps between classes, flagging loads over 18 credits
- Trained a scikit-learn linear regression on a hand-built dataset to predict course workload from credits and weekly meeting hours, penalizing overloaded schedules

Game Records Leaderboard | Java, JUnit 5 | CS 400 team project, backend developer Fall 2026
- Built the backend for a game leaderboard that loads 600 player records from CSV into a sorted tree and answers score-range queries, completion-time filters, and top-10 lowest-damage lookups
- Mapped CSV columns by header name and skipped malformed rows, so files with reordered or extra columns still load
- Wrote JUnit 5 tests against a placeholder tree, verifying backend logic before the full tree implementation was available

Red-Black Tree | Java, JUnit 5 | CS 400 coursework Fall 2026
- Implemented a generic red-black tree on my own binary search tree, rebalancing inserts with recoloring and rotations
- Tested rotations and each insert-repair case (red-uncle recolor, zig-zag, straight-line) with JUnit 5 tests

Version Control System | Java | CS 300 coursework Spring 2026
- Implemented a simplified version-control system modeling repositories, branches, commits, and version history
- Wrote CommitStack and Queue from scratch (no Java collections) alongside Branch and Repository classes

EXPERIENCE

Bakke Recreation & Wellbeing, UW–Madison | RecWell Operations Attendant | Madison, WI Aug 2026 – Present
- Run walkthroughs, enforce policies, and help patrons at a facility averaging ~4,000 daily check-ins
- Set up spaces and equipment for events and daily use, sanitize equipment, and close the facility at end of day
- Responded to a patron's volleyball injury by notifying a supervisor and following the facility's injury procedure

Bean & Creamery, UW–Madison | Barista | Madison, WI Aug 2025 – May 2026
- Served 100–150 customers per shift on the POS register while working 10–12 hours a week during the school year
- Led shifts and trained 4+ new baristas; handled opening, closing, restocking, and cleaning

1Rivet | IT Department Intern | Gujarat, India May 2024
- Supported IT development in UI/UX design, project planning, and full-stack work with Java, C++, and web frameworks

TECHNICAL SKILLS

Languages: Python, Java, HTML
Libraries & Frameworks: Flask, scikit-learn, NumPy, Jinja2
Concepts: Data structures (binary search trees, red-black trees, stacks, queues), object-oriented programming, unit testing
Tools: Git, JUnit 5, VS Code, IntelliJ IDEA

LEADERSHIP, SERVICE & AWARDS

- **Head Boy, Atul Vidyalaya:** Chosen by student-teacher vote and principal interview to lead ~1,400 students 2024
- **Cleanup Drive Organizer:** Led a cleanup in Atul and Valsad for Atul Foundation's Ujjwal Atul program 2024
- **Co-founder & President, Social Welfare Club:** Raised $1,000 in one week for underprivileged students 2024
- **Awards:** 1st, Newton Physics; 2nd, Avogadro Chemistry (Waterloo contests); Math Olympiad Rank 2 2023 – 2024
- **Athletics:** School soccer captain (Gujarat champions, 4th at CISCE nationals); national badminton (2x) 2023 – 2024
- **Workshop:** IIT Delhi Techstein workshop in AI, robotics, and Python 2024
- **Activities:** Google Developer Student Club; Badminton Club, UW–Madison 2025 – Present